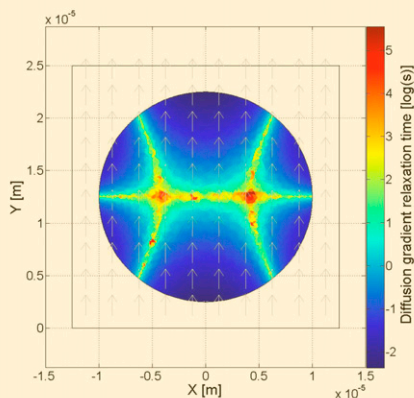


Special Section: Patterns

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To advance the fundamental understanding necessary to translate measured nuclear magnetic resonance (NMR) relaxometry data into pore-space properties, NMR signals were simulated at the microscale using a finite element analysis. Subjects studied were a pore-size- and frequency-dependent ratio of transverse and longitudinal relaxation times, coupling, and heterogeneity of surface relaxivity.

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Microscale Simulations of NMR Relaxation in Porous Media Considering Internal Field Gradients

Longitudinal and transverse nuclear magnetic resonance (NMR) relaxation signatures in porous rock were simulated on the microscale to examine and quantify how physical hydrologic parameters, such as rock-surface properties and pore sizes, affect longitudinal and transverse NMR signals of real, complex media. Parameters studied were: magnetic field strength, rock susceptibility, pore coupling, and surface reactivity. Using the finite element method (FEM), simulations of the spatial- and time-dependent magnetization evolution in arbitrary pore geometries, diffusion regimes, and heterogeneous distributions of rock surface properties, i.e., surface relaxivity, were compiled using an adapted generic diffusion model coupled with magnetic gradient field calculations. The numerical simulations were validated using analytical solutions that are available for simple pore geometries. We observed a pore-size-dependent ratio of transverse T_2 and longitudinal T_1 relaxation times, and thus a pore-size-related and rock-susceptibility-dependent effective transverse surface relaxivity was deduced. This can be used to improve estimates of pore sizes and thus of permeability from transverse NMR relaxometry measurements. Simulations of connected pore systems showed significant influences of interpore coupling at hydrologically relevant pore sizes, e.g., fine sands. Depending on the dominant diffusion regime, the typically heterogeneous distribution of surface relaxivities in rocks and sediments, i.e., geological noise, can lead to a significant underestimation of derived pore sizes and thus of permeability.

Abbreviations: DG, diffusion gradient; FEM, finite element method; FID, free induction decay; NMR, nuclear magnetic resonance.

In petrophysical applications of NMR, the measured relaxation signals originate from the fluid-filled pore space. Hence, in water-saturated rocks or sediments, the water content directly corresponds to the initial amplitude of the recorded NMR relaxation signal. The relaxation rate (longitudinal/transverse relaxation time ratio T_1/T_2) depends on the pore size and physicochemical properties of the rock–fluid interface (surface relaxivity) and on the concentration of paramagnetic ions or molecules (e.g., dissolved O_2) in the fluid phases (bulk relaxivity) (e.g., Bryar et al., 2000; Keating and Knight, 2007). Therefore, NMR can provide estimates of hydraulic parameters, i.e., pore size distribution or permeability. In this respect, there is a large variety of empirical or semiempirical relations (Kenyon et al., 1988; Dunn, 2002; Sen et al., 1990). Other relations frequently used as permeability estimators from NMR are based on geometrical considerations such as the Kozeny–Carman model (Carman, 1956, Ch. 1). These permeability estimators apply the analytical relations between pore size and NMR relaxation behavior derived by Brownstein and Tarr (1979) for single unconnected pores. These relations, however, are limited to simple, idealized pore models such as cylinders, spheres, or planar shapes. Furthermore, the influence of internal gradients on the transverse relaxation processes is usually not accounted for. As Pape et al. (2009) demonstrated for low-porosity reservoir rocks, however, considering the effects of internal gradients can significantly improve the estimation of pore sizes, and thus of permeability, from T_2 relaxation times.

To better understand the fundamental quantitative correlations between complex porous media and their respective NMR parameters such as relaxation times or surface relaxivity, it is necessary to revert to numerical simulations. This work aimed at improving the basic understanding of these parameters based on pore-scale simulations, thereby advancing the interpretation of NMR data by avoiding restrictive approximated interpretation schemes, e.g., for deriving pore size distributions, connectivity, inner surfaces, or permeability.

Nguyen and Mardon (1995) initially proposed using a FEM to simulate NMR processes at the pore scale and to eventually couple these equations with transport equations (e.g., Navier–Stokes) in a porous medium to investigate and quantify the links between NMR and permeability. Due to the limited computing capacities at that time, however, their investigations were restricted to small, simple, two-dimensional pore geometries and respective finite element meshes. In this study, we developed a multiphysics approach using a FEM to numerically simulate NMR T_1 and T_2 relaxation data at the microscale to study how physical and hydrologic parameters affect NMR relaxometry data. **The studied parameters that influence NMR signals are: internal field gradients, pore connectivities, and various diffusion regimes.** The COMSOL Multiphysics programming environment (COMSOL, Burlington, MA) was applied to simulate the arbitrary model geometries, thereby allowing a multiphysics three-dimensional modeling of the NMR behavior of the porous medium model along with the electromagnetic properties of the model. Nuclear magnetic resonance signatures can be simulated in both time and eigenvalue domains, which allows NMR relaxation times and amplitudes to be directly derived without the need for prior processing of the time-domain data, e.g., by inverse Laplace transformation. This approach thus reduces computational costs and avoids ambiguities when estimating parameters by inverse modeling.

Theory of NMR Relaxation in Porous Media

The phenomenon of NMR occurs when the nuclei of certain atoms immersed in a static magnetic field B_0 are then exposed to a second oscillating magnetic field. In hydrologic and petrophysical practice, usually the nuclear magnetization of H^1 in the pore fluid (e.g., water or oil) is relevant (Dunn, 2002; Kenyon, 1997). The relaxation of observable NMR signals in pore fluids results from the interaction of nuclear spins with their surrounding environment, i.e., the bulk fluid and pore walls.

Bloch–Torrey Equations

The evolution of magnetization during excitation and the subsequent NMR relaxation of the energized nuclei in free fluid is well described with the phenomenological Bloch–Torrey equations (Bloch, 1946; Torrey, 1956). They describe the x' , y' , and z components of the magnetization vector $\mathbf{m}$ as a function of time in the rotational frame of reference—precessing along the z axis with the angular Larmor frequency $\omega_L = 2\pi\gamma B_0$ —according to

$$\frac{\partial \mathbf{m}}{\partial t} = \gamma (\mathbf{m} \times \mathbf{B}^{\text{ex}}) - \begin{pmatrix} \mathbf{m}_{x'}/T_2 \\ \mathbf{m}_{y'}/T_2 \\ (\mathbf{m}_{z'} - \mathbf{m}_0)/T_1 \end{pmatrix} - \nabla \mathbf{D} \nabla \mathbf{m} \quad [1]$$

where $\mathbf{m} = \mathbf{m}(\mathbf{r}, t)$ denotes the magnetization density vector in the rotational frame of reference and γ is the gyromagnetic ratio of the protons. The diffusion tensor $\mathbf{D}$ simplifies to $D = \|\mathbf{D}\|$ in the case

of isotropic diffusion being assumed in the presented simulations. External magnetic fields in addition to $\mathbf{B}_0 = (0, 0, B_0)^T = \text{constant}$ are denoted by $\mathbf{B}^{\text{ex}} = [B_x, B_y, \mathbf{G}(\mathbf{r}, t)\mathbf{r} + \Delta\omega(\mathbf{r})/\gamma]^T$, where $B_1 = B_x + jB_y$ is the complex oscillating radio frequency (rf) field of the NMR excitation pulse and $\Delta\omega$ denotes the frequency offset between B_1 and ω_0 . The global external gradient field vector of the static magnetic field is given by $\mathbf{G}_k(\mathbf{r}, t) = [B_k(\mathbf{r}, t) - B_{0k}]/r_k$ with $k = x, y, z$.

These relations represent the fundamental equations of motion of the magnetic moment vector $\mathbf{m}$ by considering diffusion plus an additional sink term governing the attenuation of the system to equilibrium. The relaxation constants T_2 and T_1 reflect the homogeneous decay of the macroscopic nuclear magnetization (spin-spin or transverse relaxation time constant, T_2) and energy exchange with the surrounding environment (spin-lattice or longitudinal relaxation time constant, T_1) returning the system to equilibrium magnetization density $m_0(t=0) = m_0\mathbf{u}_z$. After termination of the rf excitation field, the external magnetic field vector simplifies to $\mathbf{B}^{\text{ex}} = [0, 0, \mathbf{G}(\mathbf{r}, t)\mathbf{r}]^T$ and Eq. [1] can be written as

$$\frac{\partial m_{\perp}}{\partial t} = -j\gamma \mathbf{r} \mathbf{G} m_{\perp} - \frac{m_{\perp}}{T_2} + \nabla \mathbf{D} \nabla m_{\perp} \quad [2a]$$

$$\frac{\partial m_z}{\partial t} = \frac{m_z - m_0}{T_1} + \nabla \mathbf{D} \nabla m_z \quad [2b]$$

where the complex transverse magnetization $m_{\perp} \equiv m_x + jm_y$ describes the free induction decay (FID) of the transverse magnetization in the observational xy plane that, in practice, is mainly dominated by signal dephasing of the total magnetization $M_{\perp}$ due to field inhomogeneities (local gradients) in the static field B_0 . To eliminate the complex dephasing term of the transverse magnetization in Eq. [2a], and thus refocusing $M_{\perp}$, a second excitation pulse shifted by 180° , i.e., $j\gamma \mathbf{r} \mathbf{G} m_{\perp}$, can be applied after a time $t = \tau$ stimulating a Hahn echo (Hahn, 1950). The repetition of such a procedure results in the well-known CPMG echo train (Carr and Purcell, 1954; Meiboom and Gill, 1958), which is commonly used in T_2 relaxation experiments. The total transverse magnetization experiences further irreversible degradation due to dephasing, however, as the molecules diffuse through the inhomogeneous magnetic field during the time interval τ . To take into account such diffusive motion of the excited spins in the presence of a magnetic gradient field during the time interval τ , an additional time-dependent sink term (irreversible) has to be incorporated into Eq. [2a], yielding

$$\frac{\partial m_{\perp}}{\partial t} = \underbrace{-j\gamma \mathbf{r} \mathbf{G}(\mathbf{r}) m_{\perp}}_{\text{reversible by Hahn echo}} - \underbrace{D[\gamma \mathbf{G}(\mathbf{r})t]^2 m_{\perp}}_{\text{irreversible (diffusion gradient)}} - \frac{m_{\perp}}{T_2} + \nabla \mathbf{D} \nabla m_{\perp} \quad [3]$$

CPMG
FID

for transverse relaxation with a diffusion gradient and FID (e.g., Dunn, 2002).

Magnetization Evolution in a Restricted Pore Space

Considering NMR relaxation in a confined system, e.g., a saturated porous medium, the relaxation rate of excited nuclei is further accelerated due to surface interactions at rock–fluid interfaces. To explain the NMR relaxation behavior in an environment where diffusion is restricted, e.g., a single isolated pore, Brownstein and Tarr (1979) introduced Neumann boundary conditions to Eq. [1] at the fluid interfaces, e.g., pore walls.

At the (active) pore surface—i.e., within a thin volume layer of thickness d in the range of a few Angstroms near the pore surface (Fig. 1)—the excited nuclei experience an accelerated surface relaxation rate $1/\tau_s$. The specific mechanism for such relaxation depends strongly on the magnetic dipolar interactions of the spins in the fluid with magnetic dipoles within this layer or at the surface of the fluid–rock interface (Torrey, 1956; Foley et al., 1996). Such interactions are very short ranged and are commonly assumed to be in the range of a few Angstroms. The strength of the NMR relaxation at the surface is then characterized by the surface relaxivity ρ_s defined as $\rho_s \equiv d(1/\tau_s - 1/T_b)$, where T_b is the relaxation time of the bulk fluid and τ_s is the surface relaxation time within the surface fluid layer of thickness d .

The dominant mechanism governing the surface relaxation processes in porous rocks can be attributed to the hyperfine interactions of the fluid nuclei with paramagnetic centers at the rock surfaces, e.g., most commonly Fe^{3+} and Mn^{2+} . Thereby, the transversal and longitudinal surface relaxation rates ($1/\tau_{s1,2}$) related to fluid nuclei aligned with paramagnetic ions at the rock surface is characterized by different multipolar relaxation mechanisms such that $\rho_{s2}/\rho_{s1} \geq 1$ with median experimental values of 1.59 (e.g., Korringa et al., 1962; Kleinberg et al., 1993). In this study, however, we have focused mainly on transverse relaxation processes enhanced by diffusion in inhomogeneous magnetic fields that arise from susceptibility contrasts between the pore fluid and rock matrix (internal gradients) and as such are not related to surface relaxation. Therefore, without loss of generality, we use $\rho_{s1} = \rho_{s2}$ as a common intrinsic rock property input parameter on the i th fraction of active NMR rock surface S_i , e.g., $\rho_{Si} = 10 \mu\text{m/s}$:

$$\begin{aligned} \rho_s m(\mathbf{r}, t) + D \hat{n}_i \nabla m(\mathbf{r}, t)|_{S_i} &= 0 \quad (\text{active}) \\ D \hat{n}_i \nabla m(\mathbf{r}, t)|_{S_i} &= 0 \quad (\text{passive}) \end{aligned} \quad [4]$$

where $\hat{n}_i$ is the surface outward normal at S_i . The surface relaxivity ρ_s is then a measure of the loss of magnetization and reflects the molecular interactions at the fluid boundary S . **In this respect, the gas–fluid interface in a partially saturated system would have to be treated as a passive boundary.** In hydrologic considerations, the corresponding fluid phase would be considered an active volume

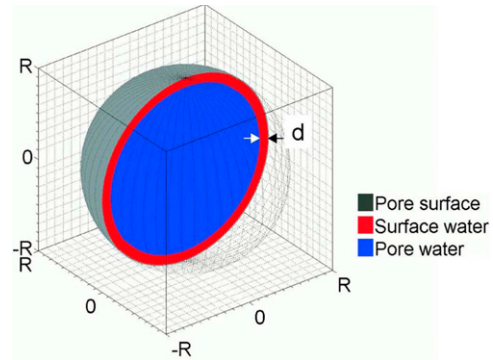


Fig. 1. Conceptual representation of volumetric water fraction and surface water fraction with thickness d of an isolated spherical pore of radius R .

(water), whereas the gas phase (air) would be a noncontributing passive volume, i.e., $m_0(t=0) = 0$. Note, that in case of an NMR active second gas or fluid phase, magnetization (spin) density $m_{0,2} \neq 0$ and diffusion coefficient D_2 would have to be set accordingly for such phase.

Summing up the introduced NMR diffusion Eq. [2b] and [3] together with the boundary conditions as given in Eq. [4] allows a comprehensive description of the evolution of magnetization in a porous medium for time-domain modeling. The formulation is valid for T_1 relaxation processes and can also be used to calculate the FID and transverse relaxation behavior in the presence of internal and external magnetic gradients.

Considering an experimentally observable scale, the spatially distributed magnetization densities m integrate to a total effective macroscopic magnetization:

$$M(t) = \int_V m(\mathbf{r}, t) d\mathbf{r} \quad [5]$$

and

$$M_0 = \int_V m(\mathbf{r}, t=0) d\mathbf{r} \quad [6]$$

where V is the total volume of the investigated pore space and M_0 is the effective magnetization in equilibrium.

In practice, e.g., to reduce computation time and to directly derive relaxation constants, it can be convenient to solve the above differential equations not in the time domain but rather in terms of their eigenvalues by separating variables according to

$$m(\mathbf{r}, t) = \sum_{n=1}^{\infty} E_n(\mathbf{r}) \exp(-t\lambda_n) \quad [7]$$

Substituting m with Eq. [7], the orthogonal eigenfunctions $E_n(\mathbf{r})$ and eigenvalues $\lambda_n \equiv 1/T_n$ basically satisfy

$$-\frac{1}{T_b} E_n + \nabla D \nabla E_n = -\lambda_n E_n \quad (T_1) \quad [8a]$$

$$-\left\{\frac{1}{T_b} - D[\gamma \mathbf{G}(\mathbf{r})t]^2\right\} E_n + \nabla D \nabla E_n \quad (T_2) \quad [8b]$$

$$-\left\{j\gamma \mathbf{r} \mathbf{G}(\mathbf{r}) + D[\gamma \mathbf{G}(\mathbf{r})t]^2 + \frac{1}{T_b}\right\} E_n + \nabla D \nabla E_n = -\lambda_n E_n \quad (\text{FID}) \quad [8c]$$

for the longitudinal and transverse NMR relaxation behavior. The boundary conditions for T_1 , T_2 , and FID are given by

$$\begin{aligned} D\hat{n}_i E_n + \rho_s E_n|_{S_i} &= 0 \\ D\hat{n}_i E_n|_{S_i} &= 0 \end{aligned} \quad [8d]$$

for active and passive boundaries, respectively. For simple geometries, e.g., spherical (Fig. 1), cylindrical, or planar shapes, Eq. [8a] can be solved analytically. From such considerations, the basic relations between NMR T_1 relaxation times and the geometrical properties of isolated pores, e.g., the pore radius, were derived by Brownstein and Tarr (1979) and are commonly used for the interpretation of NMR relaxometry measurements. They described the relaxation behavior of magnetization in surface- and diffusion-limited regimes. In the case of a surface-limited (fast) diffusion regime and assuming $ds/(v - ds) \ll 1$, a pore is characterized by a single eigenvalue λ , i.e., the NMR-labeled molecules diffuse much faster to the pore wall than the time it takes for the excited spins to decay to the equilibrium state; hence,

$$\frac{1}{T_1} = \rho_s \frac{s}{v} + \frac{1}{T_b} \quad [9a]$$

for longitudinal relaxation and

$$\frac{1}{T_2} = \frac{1}{T_2^{\text{app}}} - DG = \rho_s \frac{s}{v} + \frac{1}{T_b} \quad [9b]$$

when considering transverse relaxation times T_2^{app} measured in CPMG, where DG is the irreversible diffusion gradient relaxation rate due to magnetic gradients, and s/v represents the surface/volume ratio of the pore. Furthermore, by assuming a constant gradient field $G(\mathbf{r}, t) = \text{constant}$ and linear time steps $t = n\Delta t$ with a constant echo interval (echo time) $\Delta t = 2\tau$, Eq. [8b] can analogously be solved analytically. Thus, the transverse relaxation behavior in the presence of a linear (external) magnetic gradient field can be derived. Considering nonlinear (internal) magnetic gradients, Eq. [8b] and [8c] can only be solved numerically.

The analytical solutions of the above equations, however, are only valid under the following conditions: (i) relaxation of the nuclei begins and ends within the same pore, i.e., pores act independently; (ii) surface relaxivity is constant; and (iii) pore geometry is known and constant throughout the investigated sample. Therefore, to assess the limitations of such approximated solutions and to improve the quantitative understanding of T_1 and T_2 NMR relaxation signatures of complex porous systems (e.g., connected pores, variations of surface properties due to different rock minerals, the

impact of internal gradients on derived T_2 -based pore size distributions and permeability, partial saturation, etc.) the above NMR diffusion equations have to be solved numerically.

Calculation of the Primary Magnetic Fields (Internal Gradients)

It is well known that internal magnetic field gradients in saturated rocks originate from susceptibility contrasts between the pore fluid and the minerals in the rock matrix (e.g., Kleinberg et al., 1994; Coates et al., 1999; Godefroy et al., 2001). To model the increased decay rate observed in the transverse NMR relaxometry, i.e., CPMG echo trains and FID, due to the diffusive motion of the NMR-labeled molecules in a nonlinear magnetic gradient field (internal gradient), the spatial distribution of field gradients within the simulated fluid-matrix system can be calculated according to Ampere's law (Nabighian, 1987):

$$\nabla \times \left[\frac{1}{\mu_0 \mu_r(\mathbf{r})} \nabla \times \mathbf{A} \right] = \mathbf{J} \quad [10]$$

with

$$\mathbf{B} = \nabla \times \mathbf{A} \quad \text{and} \quad \mathbf{B} = \mu_0 \mu_r \mathbf{H}$$

and outer boundary conditions

$$\nabla \times \mathbf{H} = \nabla \times \mathbf{H}_0 \quad [11]$$

at the model limits and with inner continuity requirements

$$\nabla \times (\mathbf{H}_{S_1} - \mathbf{H}_{S_2}) = 0|_{S_{1,2}} \quad [12]$$

at interfaces $S_{1,2}$ between inner model domains such as the pore-matrix boundary. The global Larmor frequency $f_0 = \gamma B_0 = \gamma \mu_0 \mu_r H_0$ is defined by the static external primary magnetic field $\mathbf{H}_0$. The gradient in the direction of the equilibrium magnetization $B_0 = B_z$ at the point $\mathbf{r}$ is given by $\mathbf{G}(\mathbf{r}) = \partial/\partial_z \mathbf{B}(\mathbf{r})$. The terms $\mathbf{A}$, $\mathbf{H}$, and $\mathbf{B}$ denote the local magnetic potentials, fields, and flux densities, respectively; $\mu_r(\mathbf{r}) = 1 + \kappa(\mathbf{r})$ is the spatial distribution of the magnetic permeability (susceptibility κ) in the system. In this study, we set $\mu_r^{\text{fluid}} = 1$ and assumed mean values for μ_r^{rock} throughout the rock matrix and thus $\mu_r^{\text{rock}} > \mu_r^{\text{fluid/gas}}$. Dissolved ferro- or paramagnetic ions such as Fe^{3+} or Mn^{2+} , however, can significantly increase the magnetic permeability of the pore fluid. Actually, μ_r can also exhibit significant variations within a rock sample, e.g., due to individual grains or rock fractions with a high Fe content. Typically, susceptibility values for sediments and rocks range from $\kappa \leq 10^{-4}$ SI (e.g., clean sandstone or silica) up to $\kappa \geq 10^{-2}$ SI (e.g., volcanic rocks).

Numerical Realization

As the description of the evolution of magnetization in the pore fluid is very similar to classical differential equations for diffusion processes, approaches frequently used in fluid dynamic sciences can be used to solve Eq. [1–3] with boundary conditions according to

Eq. [4] in the time domain, e.g., random walk strategies, lattice Boltzmann methods (LBMs), or FEMs.

We used COMSOL Multiphysics, a commercial program based on finite elements that includes simulation toolboxes for general convection and diffusion processes in arbitrary three-dimensional geometries. Optionally, these models can be linked to other physical simulations, such as corresponding magnetic field calculations (internal gradients), fluid flow, or heat transfer. In the FEM, the geometric model is divided into mesh elements (Fig. 2) and the above differential equations are solved for magnetization densities $m_i(t)$ at each mesh element i of the model (Zienkiewicz, 2005).

In contrast to random walk methods or LBMs, the FEM approach permits the NMR diffusion equations to be solved in their eigenvalue domain, thus significantly reducing the computational cost. Consequently, arbitrary pore geometries can be simulated and a direct derivation of NMR relaxation times (eigenvalues) can be derived directly without the need for consecutive inverse modeling in the time domain, e.g., by inverse Laplace transformation.

In this study, we considered only the evolution of magnetization densities of the NMR-labeled H_2O molecules in the saturated pores, i.e., initially $m(x,y,z,t=0) = M_0/V$. The simulations were compiled using a medium performance dual core personal computer with 2Gb of RAM. Depending on the model design, the computational time was generally in the range of several seconds up to a few minutes.

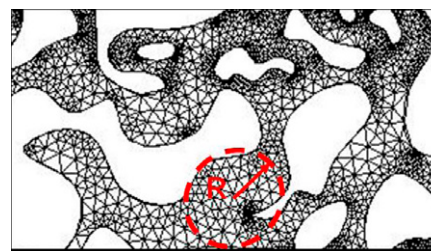


Fig. 2. Schematic cross-section of a porous medium with finite element mesh of the pore space and approximated pore radius R of a single pore body.

Results and Discussion

Model Validation Using Analytical Solutions

To validate the simulations, we compared the numerical compilations with analytical solutions that are available for isolated pores with simple geometries. Simulation results for an isolated spherical pore in surface-dominated (one dominant decay time per eigenvalue) and diffusion-dominated (multiple decay times per eigenvalue) regimes using 4166 mesh elements are in a very good agreement with the analytical results with deviations of $<0.1\%$ (Fig. 3). In comparison, Nguyen and Mardon (1995) used an axial symmetry with eight nodes in their initial models.

To validate our simulations with respect to transverse NMR relaxation in a gradient field as stated in Eq. [3] and [8b], we simulated

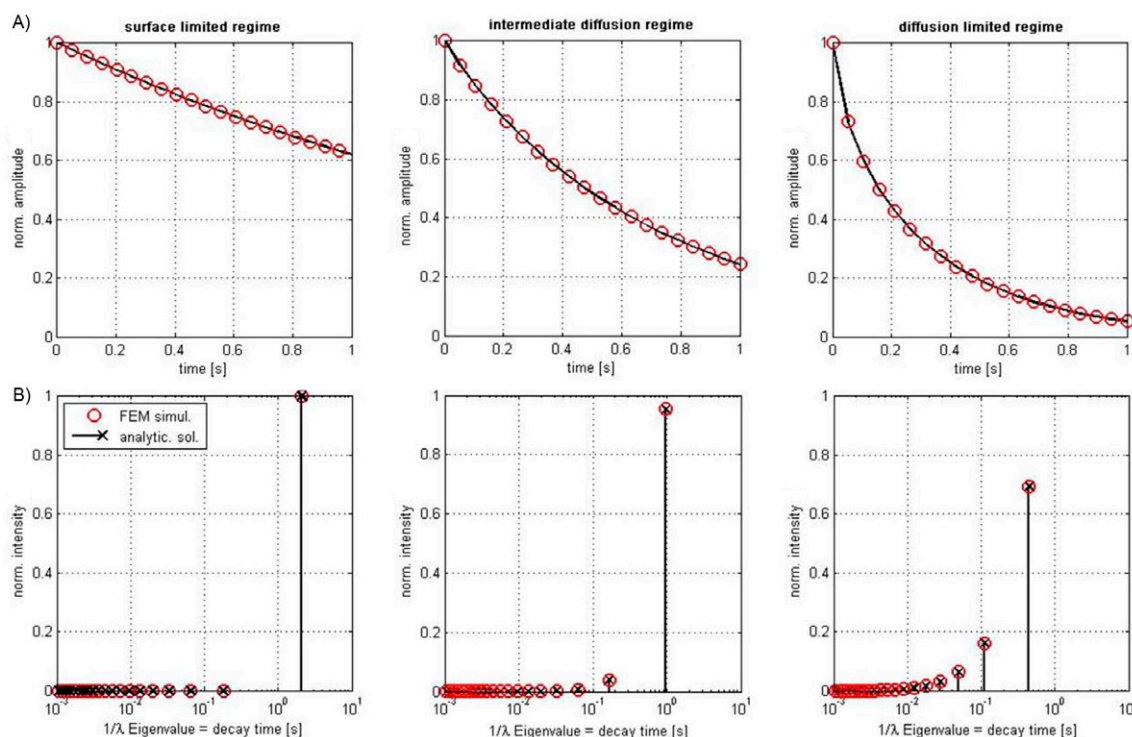


Fig. 3. Model validation of numerical simulations (finite element method [FEM]) of NMR relaxation in (A) time and (B) eigenvalue domains with analytical solutions (crosses) for a single saturated spherical pore of radius $R = 40 \mu m$ (diffusion coefficient of water $D = 2.5 \times 10^{-9} m^2/s$) in fast (surface relaxivity $\rho_s = 1 \mu m/s$), intermediate ($\rho_s = 15 \mu m/s$), and diffusion-limited regimes ($\rho_s = 75 \mu m/s$).

an unrestricted relaxation environment in free water with an intrinsic bulk relaxation time $T_b = 3$ s. The simulations used constant gradients $G(\mathbf{r}, t) = G = \text{constant}$ throughout the model, with a zero surface sink term at the model boundaries, i.e., $\rho_s = 0$. In Fig. 4, the square of the magnetic field gradient is plotted vs. effective transverse relaxation times at different CPMG echo spacings for analytical (lines) and numerical (symbols) calculations. The results are in good agreement. As DG relaxation becomes the dominant term in Eq. [3] for increasing echo times 2τ and field gradients G , the slope of each curve becomes proportional to the diffusion constant of the fluid.

Frequency Dependence of Transverse Relaxation Time Due to Internal Gradients

In NMR applications, the primary static magnetic field B_0 —usually generated by strong permanent or electro-magnets—defines the resonance frequency (Larmor frequency f_l) of the investigated nuclei, e.g., ^1H isotopes. Typically, Larmor frequencies used in petrophysical NMR relaxometry range from kilohertz (e.g., the Earth's magnetic field: $B_0 \sim 30\text{--}50 \mu\text{T}$) up to a few 10s of megahertz (e.g., MRI: $B_0 \geq 0.25$ T). The resulting internal magnetic field gradients in rocks stem from susceptibility contrasts between the pore fluids and rock matrix or are due to minerals with different susceptibilities. Such internal gradients naturally increase as the field strength of the magnetic field increases, and thus an enhanced observed CPMG relaxation rate $1/T_2$ will be observed at higher Larmor frequencies. The dependence of the transverse NMR relaxation time on the Larmor frequency as a function of susceptibility contrast between pore water and rock matrix in a saturated porous rock were simulated for the model of a single capillary pore (radius $R = 1 \mu\text{m}$, $\rho_s = 10 \mu\text{m/s}$) for a frequency range of $f_l = 10^3$ to 10^7 Hz (Fig. 5).

The simulations were compiled for a susceptibility range of $\kappa = 10^{-4}$ SI (e.g., clean sandstone) up to $\kappa = 0.1$ SI (e.g., volcanic rocks). The simulated NMR relaxation times show the strong increase in relaxation rates with increasing field strength of B_0 and thus reflect the impact of internal gradients on the resulting CPMG relaxation measurements in petrophysical NMR. Note that at very low B_0 fields, e.g., with Larmor frequencies in the kilohertz range or less, the influence of the gradient fields becomes less dominant and eventually T_2 draws near to the respective T_1 relaxation times, even for the high κ contrasts utilized in this model. (In low magnetic fields and negligible internal gradients, $T_2 \sim T_1$). Therefore, NMR relaxometry experiments at lower magnetic fields, i.e., low-field NMR, yield a more accurate estimate of petrophysical properties from T_2 relaxation times, with a tradeoff, however, toward lower signal amplitudes as the spin magnetization scales with magnetic field intensity. On the basis of such numerical case studies, the dependency of T_2 on both rock susceptibility and Larmor frequency can be used to quantify and compensate for such interfering influences of high susceptibility contrasts when deriving pore size distributions from NMR relaxometry.

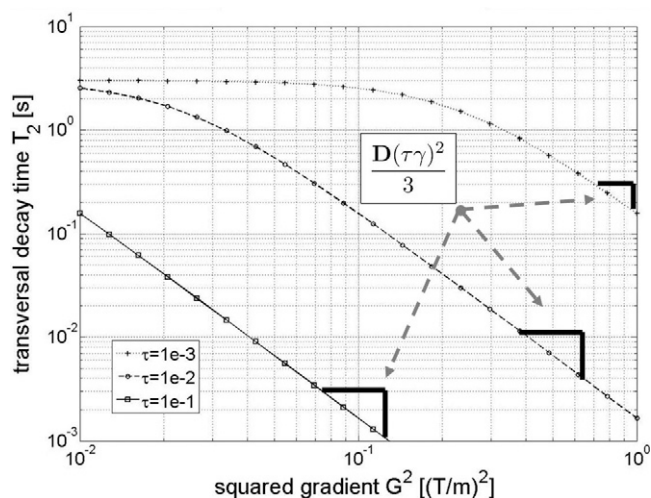


Fig. 4. Finite element method modeling (symbols) and analytical solution (lines) for increased NMR relaxation rates due to molecular diffusion in a constant magnetic gradient field for bulk water (water diffusion coefficient $D = 2.5 \times 10^{-9} \text{ m}^2/\text{s}$) at different echo times τ . The slope at high gradients is directly proportional to D (γ is the gyro-magnetic ratio of the protons).

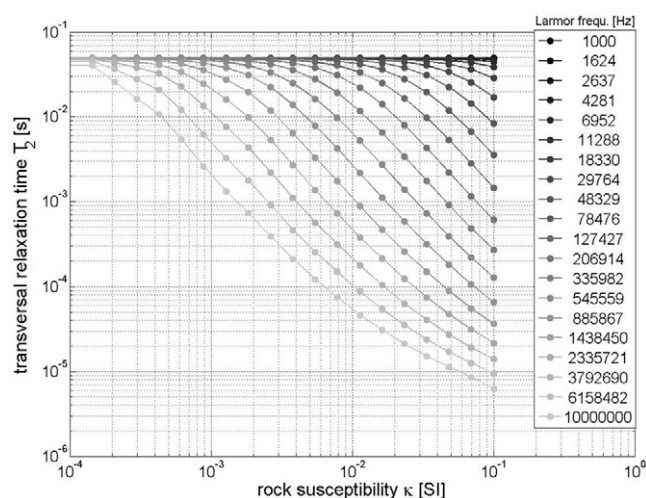


Fig. 5. Nuclear magnetic resonance transverse relaxation times of an isolated capillary pore (radius $R = 1 \mu\text{m}$, surface relaxivity $\rho_s = 10 \mu\text{m/s}$) as a function of matrix susceptibility in the presence of internal gradients. Simulations were compiled for Larmor frequencies in the range 1 kHz (black) up to 10 MHz (light gray).

Enhanced Transverse Relaxation Time as a Function of Pore Size in the Presence of Internal Gradients

For Larmor frequencies $f_l \leq 4$ MHz—which is generally considered low-field NMR—the diffusion effects for T_2 can be minimized by using a sufficiently small echo spacing τ in a CPMG experiment. Then T_2 can be used as a T_1 proxy to derive pore size distributions. In this respect, a constant T_2/T_1 ratio for all pore sizes is assumed (e.g., Kleinberg et al., 1993).

In the pore space, however, internal magnetic field gradients, and thus the DG, are at a maximum near the pore–rock interfaces, i.e., where a high contrast in κ occurs. In Fig. 6, the spatial distribution of DG within a saturated capillary pore for a matrix susceptibility $\kappa_{\text{matrix}} = 10^{-2}$ SI, echo spacing $\tau = 0.5$ ms, and Larmor frequency $f_1 = 2$ MHz is plotted. The relaxation times generally decrease toward the pore walls (dark blue), which significantly affects the resulting effective transverse relaxation time of this pore, i.e., $T_2 = 0.5$ s (no gradient) and $T_2 = 5.5 \times 10^{-2}$ s (with internal gradients). Following the above considerations in a small pore, a larger fraction of its total NMR-labeled volume is affected by the gradient field, and thus one can assume a higher impact of the diffusion gradient on T_2 NMR signatures that originate from the smaller pores.

To study such a dependence, we simulated longitudinal (T_1) and transverse (T_2) NMR relaxation signatures of saturated capillary pores in the presence of internal gradients. The simulations presented in Fig. 7 were calculated for capillary pore geometries using a global Larmor frequency of 2 MHz and (frequently used) CPMG echo times $\tau = 0.5$ ms. The T_2/T_1 ratios generally show a strong dependence on the pore radius that becomes more distinct at low surface relaxivities ρ_s at the walls. The T_2/T_1 ratios are more strongly affected at low ρ_s because of the long relaxation times for low ρ_s . In contrast, the short relaxation times at high ρ_s cannot be reduced much further by the diffusion gradient. Actually, as surface relaxivity usually reflects paramagnetic substances at the pore wall, e.g., hematite, an additional increase in ρ_s would be expected with increasing internal gradients in a rock sample due to an increased surface relaxation rate $1/\tau_s$. Even so, the results plotted in Fig. 7 support a strong relation between pore size and the T_2/T_1 ratio.

As a consequence of an increase in the rock's susceptibility, the resulting effective transverse relaxation times decline. As depicted in Fig. 8, such a decrease in T_2 strongly depends on the respective pore size. The simulations show a significant impact on the resulting relaxation rates at $\kappa \sim 10^{-2}$ SI for the large pores ($R = 100$ μm). A significant decrease in effective transverse relaxation for the small pore sizes ($R = 1$ μm) occurs already at $\kappa \sim 10^{-3}$ SI. In practice, it is commonly assumed that ρ_s does not vary significantly throughout the medium, i.e., $\rho_s = \text{constant}$ for all signal-contributing pores, although there is no direct experimental justification. From Eq. [9a] and [9b] it follows that the surface relaxivity for a single pore is given by

$$\rho_{s_k} = \frac{T_b - T_k}{T_b T_k} \frac{v}{s} \quad \text{with } k = 1, 2 \quad [13]$$

where v/s is the (active) volume to surface ratio of the pore. For a pore space consisting of several pore sizes, i.e., integrating over the entire pore space, Eq. [13] extends to

$$\rho_{s_k} = \frac{V}{S} \frac{T_b - \hat{T}_k}{T_b \hat{T}_k} \quad \text{with } \hat{T}_k = \sum_{i=1}^N \frac{I_i}{T_i} \text{ and } \sum_{i=1}^N I_i = 1 \quad [14]$$

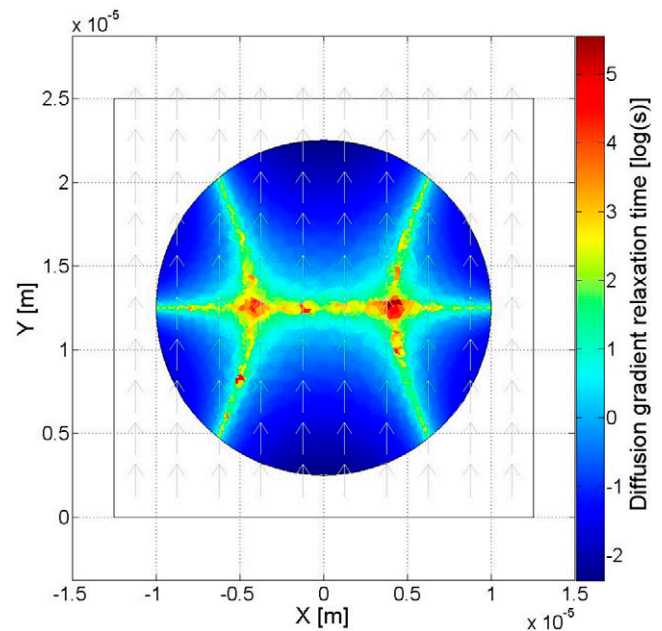


Fig. 6. Example of the spatial distribution of the diffusion gradient term DG in a capillary pore. Model parameters: Larmor frequency $f_1 = 2$ MHz, matrix susceptibility $\kappa = 10^{-2}$ SI. Arrows show the direction of the static magnetic field B_0 .

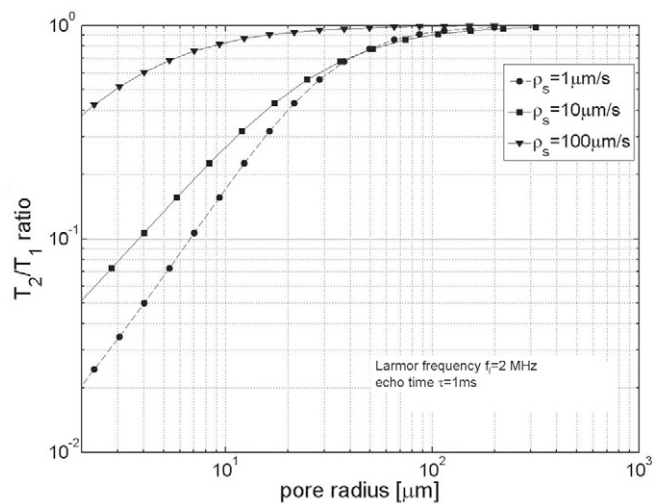


Fig. 7. Ratios of transverse to longitudinal NMR relaxation times T_2/T_1 of saturated isolated capillary pores in the presence of internal gradients as a function of the pore radius (model parameters: surface relaxivity $\rho_s = 1, 10$, and 100 $\mu\text{m/s}$, Larmor frequency $f_1 = 2$ MHz, matrix susceptibility $\kappa = 0.1$ SI).

where $\hat{T}_k$ is a weighted relaxation time, I_i and T_i are the relative signal contribution and respective relaxation time of the i th pore of the system (note that this relation is strictly only valid in a surface-limited diffusion regime), and $V = \sum v_i$ and $S = \sum s_i$ are the total volume and surface of the system, respectively.

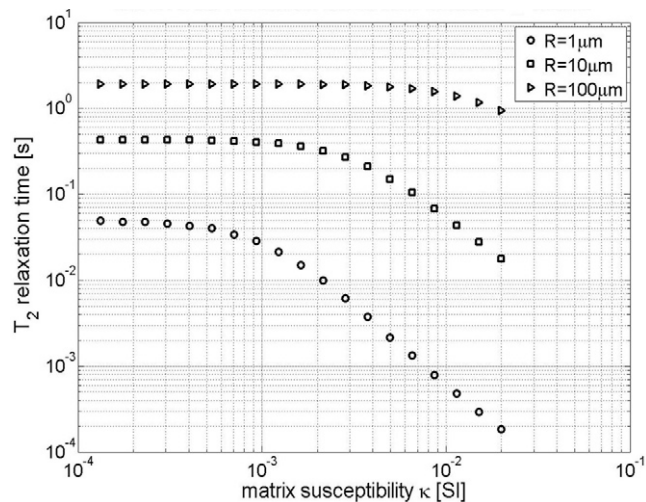


Fig. 8. Effective transverse relaxation times in the presence of internal gradients for different pore radii R as a function of rock susceptibility (model parameters: surface relaxivity $\rho_s = 10 \mu\text{m/s}$, Larmor frequency $f_l = 2 \text{ MHz}$, echo time $\tau = 0.5 \text{ ms}$).

For the simulations shown in Fig. 8 to 10, a simple capillary pore with radius R was used, thus $v/s = R/2$. Following from Eq. [14] with a known volume/surface ratio of the pore model, we can derive a pore-size-specific enhanced transverse surface relaxivity parameter ρ_{2s} that increases with increasing matrix susceptibility (Fig. 9). The parameter $\rho_{2s}(\kappa, R, f_l)$ is then also a function of the matrix susceptibility and Larmor frequency as well as the pore size and reflects the enhanced transverse relaxation rate of the pores due to the internal magnetic field gradients in the porous medium.

As illustrated in Fig. 10, on using a constant surface relaxivity value (diamonds), i.e., $\rho_{2s} = 10 \mu\text{m/s}$, to estimate the respective pore sizes, the pore radii will be significantly underestimated when the susceptibility of the matrix increases beyond a pore-size specific threshold. This susceptibility threshold decreases with decreasing pore radii, and thus the size of small pores will be more strongly underestimated than large pores. In contrast, when using an enhanced surface relaxivity model that takes into account the rock susceptibility, the pore radii are estimated correctly (dots). The slight increase at high susceptibilities is due to the beginning of the transition from a surface-limited to a diffusion-limited regime of the relaxation processes, i.e., $\rho_{2s}R/D > 1$. Generally, this susceptibility threshold will shift to smaller values for NMR experiments at higher Larmor frequencies and vice versa. These considerations concur with observations from laboratory studies on low-porosity reservoir rocks (Pape et al., 2009). In practice, to estimate the pore-size-dependent effective surface relaxivity by inverse modeling, the susceptibility and porosity (e.g., NMR porosity) of the rock sample need to be independently measured. In this context, developing an enhanced inverse modeling based on the above simulations is still an objective of ongoing research.

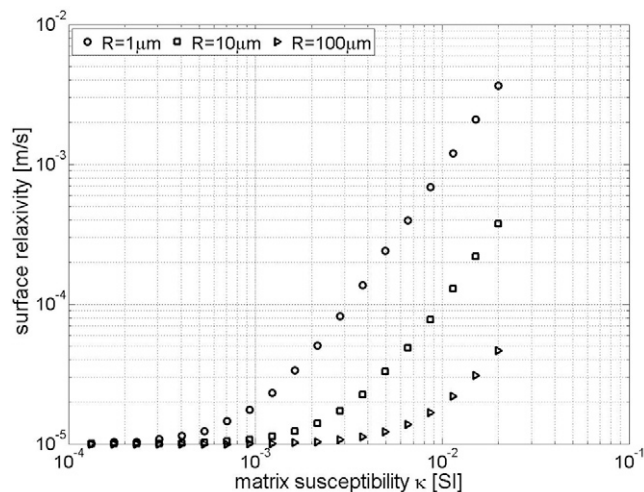


Fig. 9. Derived effective transverse surface relaxivities ρ_{2s} for different pore radii R as a function of rock susceptibility (model parameters: surface relaxivity $\rho_s = 10 \mu\text{m/s}$, Larmor frequency $f_l = 2 \text{ MHz}$).

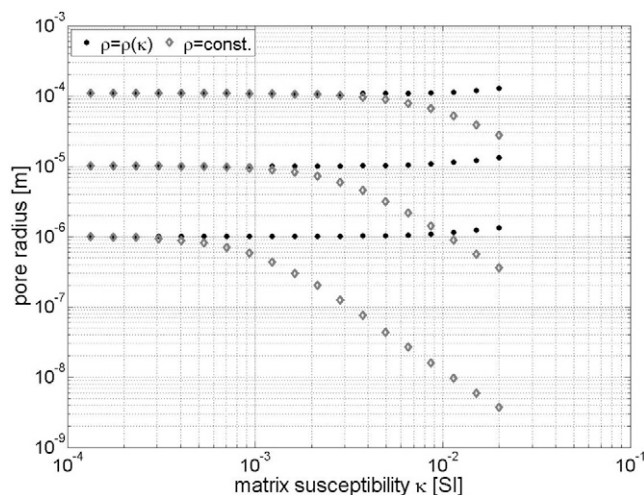


Fig. 10. Estimated pore radii from T_2 relaxation times using constant (diamonds) and effective, rock-susceptibility-dependent (dots) surface relaxivity estimates (model parameters: surface relaxivity $\rho_s = 10 \mu\text{m/s}$, Larmor frequency $f_l = 2 \text{ MHz}$, pore radii $R = 1, 10, \text{ and } 100 \mu\text{m}$).

Pore Coupling

Common quantitative interpretation of NMR relaxometry data in porous media uses the classical relation of NMR relaxation times and pore sizes for single isolated pores as derived by Brownstein and Tarr (1979). Because the pores are usually connected in porous rocks, the concept of separated individual pores is limited for the investigation of coupling between pores, i.e., the diffusive motion of NMR-labeled molecules between pores. Due to the diffusion constant of water, i.e., the diffusion length, however, pore coupling effects become more significant in NMR relaxometry in rocks with small pore sizes, e.g., in low-porosity reservoir rocks, shales, or clay.

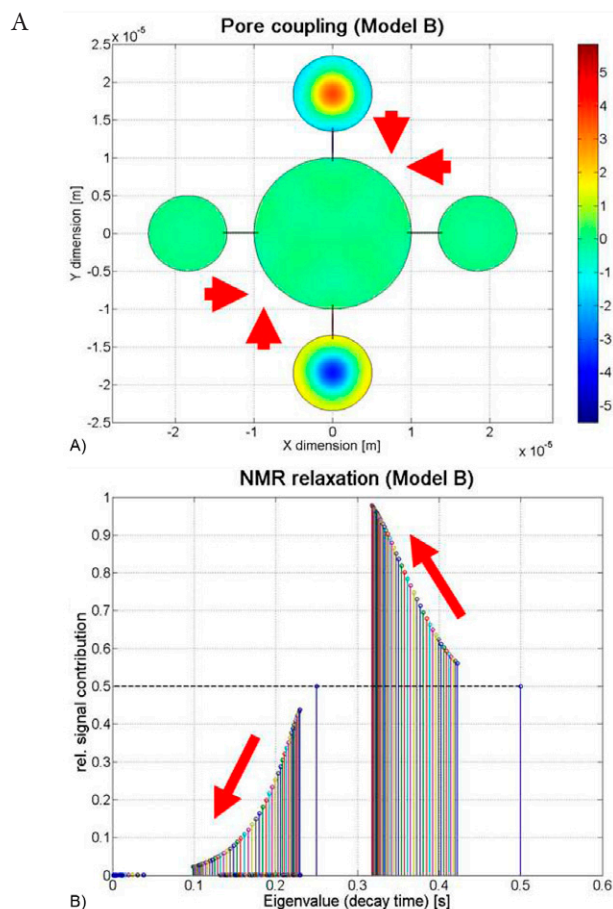


Fig. 11. (A) Model of a coupled pore system of a large pore (radius $R = 10 \mu\text{m}$) and four small pores ($R = 5 \mu\text{m}$), where the colors illustrate an example eigenfunctions $\log[E_n(r)]$ of the simulated data; and (B) simulation of longitudinal relaxation time T_1 distributions for throat lengths ranging from 0.1 to $80 \mu\text{m}$ (model parameters: surface relaxivity $\rho_s = 10 \mu\text{m/s}$, throat width $0.1 \mu\text{m}$). The arrows indicate the corresponding direction of change with decreasing throat length.

simple model of a capillary pore ($R = 10 \mu\text{m}$) that is connected to four smaller pores ($R = 5 \mu\text{m}$) by thin channels (pore throats) with widths $10^{-2} \mu\text{m}$ is shown in Fig. 11A. The total volume of the four small pores is equal to the volume of the large pore; thus we obtain equal amplitude fractions for each pore size. Simulations of longitudinal relaxation (T_1) in terms of eigenvalues were performed for different throat lengths and widths of this coupled system. Figure 11B shows the respective T_1 relaxation times and normalized amplitudes for throat lengths ranging from 0.1 to $80 \mu\text{m}$. The arrows indicate the corresponding direction of change with decreasing throat length.

The coupled system is characterized by two relaxation times that each decrease as coupling increases (red arrows). For high throat lengths (weak coupling), no significant diffusive exchange between pores takes place and the pores can be considered to be isolated, with relaxation times of 0.25 and 0.5 s. The same relative amplitudes of 0.5 for each pore size reflect the equal pore volumes of large and small pores. When decreasing the throat length and thus

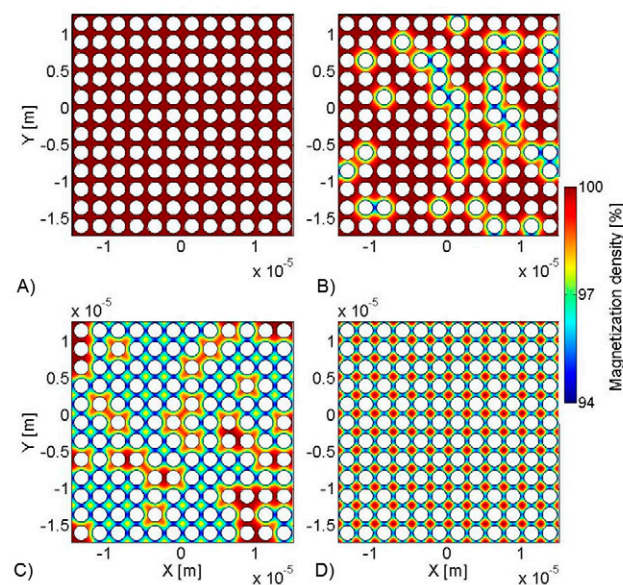


Fig. 12. Sections of a model box with 144 grains with randomly assigned low ($1 \mu\text{m/s}$) and high ($50 \mu\text{m/s}$) surface relaxivities for model realizations with (A) 0%, (B) 25%, (C) 75%, and (D) 100% high-relaxivity grain surfaces. The color code reflects the spatial distribution of the pore fluid's transverse magnetization density after the first Carr–Purcell–Meiboom–Gill (CPMG) echo ($\tau = 0.5 \text{ ms}$).

increasing the coupling strength, however, the fully coupled system will eventually be characterized by a single effective relaxation constant and a respective normalized signal amplitude of 1. Because molecular exchange by diffusion takes place between the pores, the pore sizes estimated by the classical approach fail to describe this coupled system. Note that as the dominant relaxation time of the system converges to an effective value, the lower part of the spectrum decreases along with its loss in signal strength. These implications, relevant for the NMR relaxation behavior of coupled pore systems at the pore scale, agree with the results for porous medium models obtained by McCall et al. (1991).

Heterogeneity of Surface Relaxivities

Porous rocks and sediments usually are composed of fractions of different mineral grains such as quartz, clay, or hematite. Naturally such minerals can have different properties with respect to their paramagnetic ion content at the inner surfaces and thus show highly different mineral-specific surface relaxivities, i.e., $\rho_s^{\text{quartz}} \ll \rho_s^{\text{hematite}}$.

In practice, to estimate pore sizes from NMR relaxometry relaxation data, effective mean values of ρ_s are derived with uncertainties due to measurement errors. Additionally, there will also be deviations caused by the mineral grain distributions within the rock matrix. Such natural noise usually cannot be accounted for, and information with respect to its influence and significance on the derivation of pore parameters is not well known.

To assess such influences for fast- and slow-diffusion NMR relaxation regimes, we set up a simple model box (Fig. 12) of 144

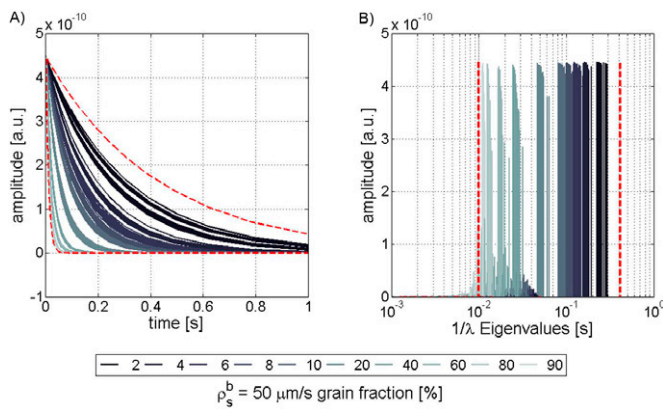


Fig. 13. Simulations with randomly distributed low ($1 \mu\text{m/s}$) and high ($50 \mu\text{m/s}$) grain surface relaxivities with 10 to 90% high-relaxivity (ρ_s^b) grain content, each with 50 random model realizations: (A) transverse relaxation curves; (B) eigenvalues (= decay times). Dashed lines show lower and upper boundaries for homogeneous low and high surface relaxivities.

spherical mineral grains (grain radii $R = 1$ and $10 \mu\text{m}$ for fast and slow diffusion, respectively), which were assigned low ($\rho_s^a = 1 \mu\text{m/s}$) and high (fast diffusion models: $\rho_s^b = 50 \mu\text{m/s}$; slow diffusion models: $\rho_s^b = 100 \mu\text{m/s}$) surface relaxivities, representing clean (e.g., quartz) and impure (e.g., hematite) mineral grains. These two surface properties were assigned randomly to the grains in the models. Figures 12A to 12D illustrate examples of random realizations (fast-diffusion model) of transverse magnetization density at the first CPMG echo ($\tau = 0.5 \text{ ms}$) for different fractions (0, 25, 75, and 100%) of high-relaxivity grains. Simulated data (fast-diffusion regime) of 50 random realizations for different fractions (0–100%) of high-relaxivity grains in the model are plotted in Fig. 13A (time domain) and 13B (eigenvalue domain). Dashed lines represent the limits for 0 and 100% high-relaxivity grain contents. Figures 14A to 14C illustrate three excerpts for high-relaxivity grain contents of 6, 20, and 60%. Here, a bimodal eigenvalue distribution at low ρ_s^b grain contents, i.e., isolated occurrences of ρ_s^b grains, converges to a weighted mean monomodal relaxation behavior with increasing high-relaxivity grain contents. Similar simulations compiled for diffusion-limited relaxation, i.e., grain sizes are upscaled by a factor of 10 and $\rho_s^b = 100 \mu\text{m/s}$, are plotted in Fig. 15. Here, at 0% high-relaxivity grain content, the relaxation is purely surface limited. Upon introducing high-relaxivity grain fractions to the model, the relaxation behavior becomes increasingly diffusion limited, and thus the system becomes characterized by an ensemble of relaxation times (e.g., Fig. 16, bottom). Upon increasing the fraction of high-relaxivity grains, the characteristic eigenvalues λ of the system increase along with an increased scattering of mean relaxation times and subsequently derived surface relaxivities for the 50 model realizations (indices 1–50). As expected, the scattering appears generally stronger in the surface-limited diffusion environment. If the model's total volume and surface area are known, Eq. [14] can be used to estimate the

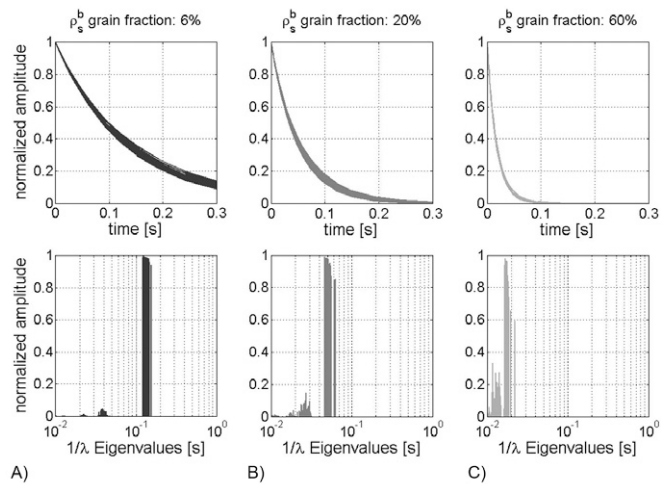


Fig. 14. Fast diffusion: NMR transverse relaxation simulations in the time domain (top) and eigenvalue (relaxation constant) domain (bottom) for 6, 20, and 60% high surface relaxivity (ρ_s^b) grain content (excerpts from Fig. 13).

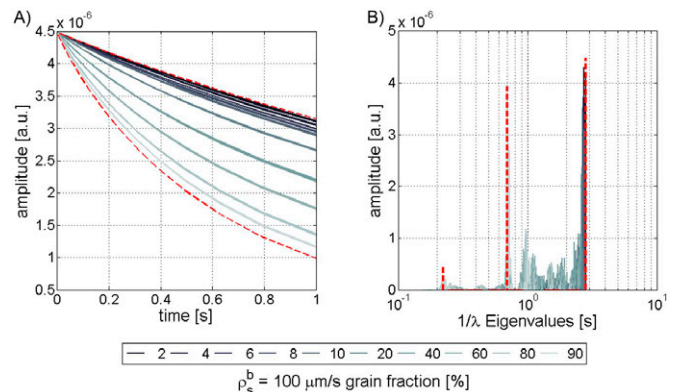


Fig. 15. Slow diffusion: simulations with randomly distributed low ($1 \mu\text{m/s}$) and high ($50 \mu\text{m/s}$) grain surface relaxivities with 10 to 90% high-relaxivity (ρ_s^b) grain content, each with 50 random model realizations: (A) transverse relaxation curves; (B) eigenvalues (= decay times). Dashed lines show lower and upper boundaries for homogeneous low and high surface relaxivities.

surface relaxivities and thus derive pore sizes from the simulated relaxation data (Fig. 17 and 18). The effective surface relaxivity of the system increases exponentially with increasing content of ρ_s^b grain surfaces.

The derived effective pore sizes for the fast-diffusion model, i.e., volume/surface ratios, were generally found to be in very good agreement of $<1\%$ with the model. In the case of intermediate to slow diffusion, however, the derived mean surface relaxivities were typically underestimated due to the resulting faster effective relaxation rates. Consequently, the deduced pore sizes were underestimated, exhibiting a maximum deviation ($\sim 15\%$) from the model at 20% high-susceptibility grain fraction. This maximum deviation can be attributed to a stronger impact from clustering

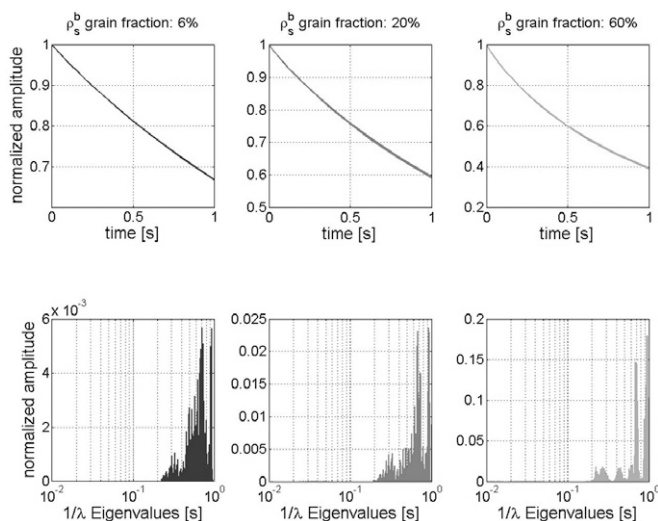


Fig. 16. Slow diffusion: NMR transverse relaxation simulations in the time domain (top) and eigenvalue (relaxation constant) domain (bottom) for 6, 20, and 60% high surface relaxivity (ρ_s^b) grain content (excerpts from Fig. 15).

(e.g., compare Fig. 12B and 12C), which then promotes the development of two distinct slow and fast relaxation regimes within the model. Furthermore, coupling between pore fractions of the model will also contribute, however, being of only minor significance in the presented models.

Conclusion and Outlook

Pore-scale simulations of longitudinal and transverse NMR relaxation signatures, taking into account the enhanced T_2 relaxation rates that arise in the presence of internal magnetic gradients in porous rocks and sediments, were implemented in a multiphysics finite elements program. The simulations can be compiled for arbitrary pore geometries with spatial distributions of surface and magnetic properties of the fluid and rock, and thus allow the fundamental NMR relaxation behavior to be studied in complex pore systems, e.g., derived from computed tomography. **The finite element approach permits the NMR diffusion equations to be solved in their eigenvalue domain and thus the NMR relaxation times (eigenvalues) to be directly derived.** The computational cost can thereby be significantly reduced and simulations can be compiled in two and also in three dimensions using a standard personal computer.

Simulations of transverse NMR relaxometry data support previous observations (e.g., Pape et al., 2009) indicating that the use of T_2 distributions as a permeability estimator tends to overestimate the contribution of smaller pores and thus to underestimate the rock permeability. To compensate for this drawback, the concept of an enhanced transverse surface relaxivity parameter that depends on pore size and rock susceptibility was introduced.

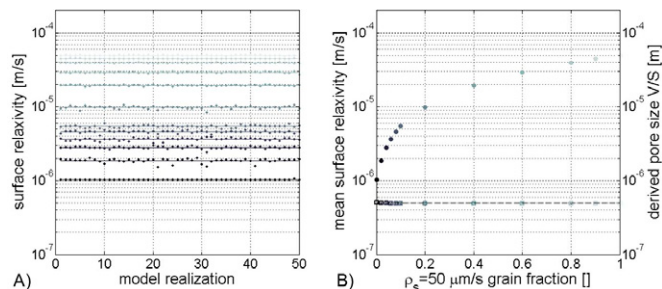


Fig. 17. Fast diffusion: derived surface relaxivities and inner surfaces for 50 realizations of random distributions of low (1 $\mu\text{m/s}$) and high (50 $\mu\text{m/s}$) surface relaxivity pairs ($\rho_s^{a,b}$) in the model box (Fig. 12): (A) effective relaxivities (dots) derived according to Eq. [14] (lines); (B) high surface relaxivity grain fraction vs. mean surface relaxivity (dots) and consecutively derived pore sizes volume/surface area (v/s , squares).

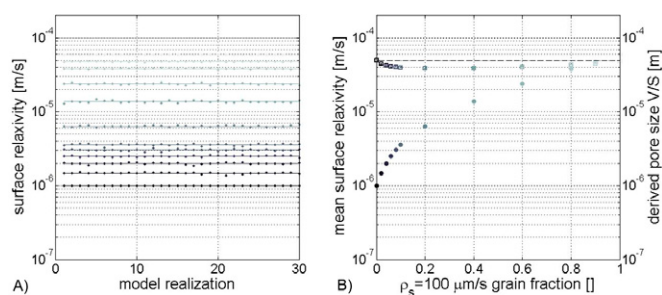


Fig. 18. Slow diffusion: derived surface relaxivities and pore size for 50 realizations of random distributions of low (1 $\mu\text{m/s}$) and high (100 $\mu\text{m/s}$) surface relaxivity pairs ($\rho_s^{a,b}$) in the model box (Fig. 12): (A) effective relaxivities (dots) derived according to Eq. [14] (lines); (B) high surface relaxivity grain fraction vs. mean surface relaxivity (dots) and consecutively derived pore sizes volume/surface area (v/s , squares).

Modeling of the transverse relaxation behavior reflects the frequency dependence of internal gradients in the rocks, which, in principle, could be used to derive effective surface relaxivities by inverse modeling or enhancing existing (semi)empirical models. Therefore, it seems promising to apply such an approach to data obtained from frequency-dependent NMR relaxometry data, e.g., utilizing transverse NMR data recorded at two different Larmor frequencies, to better estimate pore-size distributions and permeability. In this context, from the observed shifts of the transverse relaxation times at different frequencies, it seems promising to quantify frequency- and pore-size-related effective surface relaxivities and thus derive more reliable permeability estimates, in particular, for low-porosity reservoir rocks, shales, or clay. **Direct simulations of connected pores at the pore scale confirm an averaging of decay times along with an overall shift to higher decay rates related to the coupling strengths within the pore system.** Simulated NMR signatures of random grain mixtures indicated that, depending on the dominant diffusion regime, the commonly heterogeneous distribution of surface relaxivities in rocks and sediments, i.e., geological noise, can lead to a significant underestimation of derived pore sizes and thus of permeability.

On the basis of such pore-scale simulations, we aim to implement an NMR interpretation scheme for T_2 relaxometry data that will take into account pore-size-dependent transverse relaxation rates due to susceptibility contrasts between the rock and pore fluid. Using the presented finite element approach, simulations can be readily compiled for multiphase systems and thus will also be used in further studies to correlate the NMR properties and hydrologic parameters of partially saturated rocks, e.g., to derive the relative permeability and to study the transport properties of film flow. In a next step to further investigate and quantify the relationship between NMR and the transport properties of porous media, we will jointly simulate NMR and fluid flow (e.g., Navier–Stokes).

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